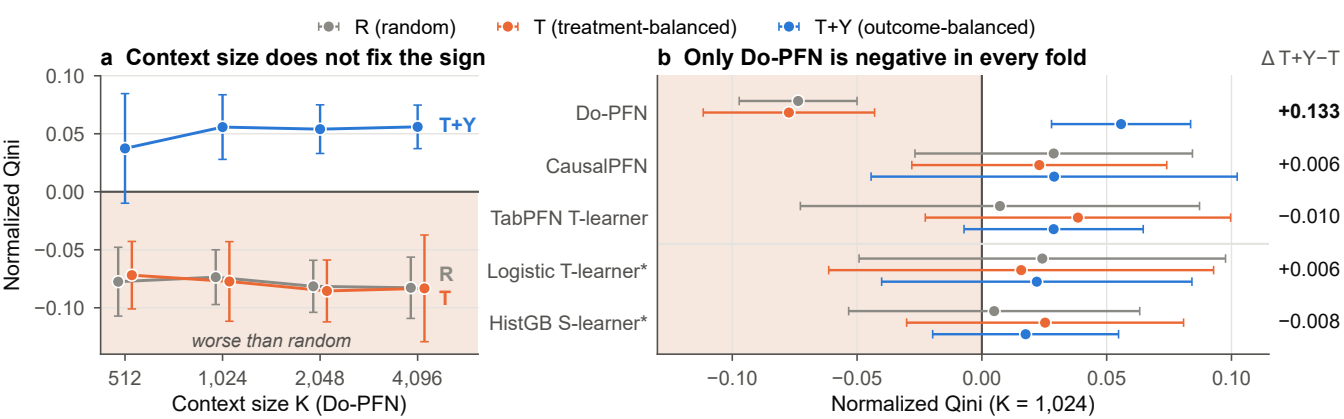

R (random)
T (treatment-balanced)
T+Y (outcome-balanced)
a Context size does not fix the sign
Normalized Qini
0.10
0.05
0.00
−0.05
−0.10
T+Y
R
T
worse than random
512
1,024
2,048
4,096
Context size K (Do-PFN)
b Only Do-PFN is negative in every fold
Δ T+Y−T
Do-PFN
+0.133
CausalPFN
+0.006
TabPFN T-learner
−0.010
Logistic T-learner*
+0.006
HistGB S-learner*
−0.008
−0.10
−0.05
0.00
0.05
0.10
Normalized Qini (K = 1,024)